PROJECT REPORT FOR

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

DIGITAL WORK PASSPORT

A PROJECT REPORT

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October, 2026

BONAFIDE CERTIFICATE

Certified that this project report **"DIGITAL WORK PASSPORT"** is the bonafide work of "HARISHNI.B (240701176) AND HARINI NAGARAJAN (240701167)" who carried out the project work under my supervision.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
1	INTRODUCTION	1
2	LITERATURE REVIEW	3
3	PROPOSED SYSTEM	4
	3.1 ARCHITECTURE DIAGRAM	
4	MODULES DESCRIPTION	7
	4.1 WORKER REGISTRATION & VOICE INTERFACE	
	4.2 WORK VERIFICATION & RATINGS	
	4.3 TRUST SCORE & INCOME RECORD	
	4.4 ACCESS & RECOMMENDATION SERVICES	
5	IMPLEMENTATION AND RESULTS	11
	5.1 EXPERIMENTAL SETUP	
	5.2 RESULTS	
6	CONCLUSION AND FUTURE WORK	15
7	REFERENCES	16

ABSTRACT

Informal workers such as domestic helpers, beauticians, tailors, cooks, vendors, and other service workers contribute significantly to everyday economic activities. However, much of their work is not formally documented. The absence of a verified work history makes it difficult for them to prove their experience, demonstrate income, obtain financial support, access suitable government welfare schemes, and find better employment opportunities.

This project proposes a **Digital Work Passport** that creates a verified digital record of informal work. The system provides a simple, voice-enabled website with regional-language support. Workers can create their profiles and record their work activities. Employers or clients can confirm completed jobs and provide ratings.

The verified work records are stored as a digital work history. Based on completed jobs, ratings, work continuity, and recorded income, the system generates a **Trust Score** and maintains an **Income Record**. These records can then support job recommendations and provide additional information for partner financial institutions, insurance providers, and government-scheme information services. The proposed system combines artificial intelligence, digital work verification, recommendation systems, voice interaction, and secure data management to improve financial and employment opportunities for informal workers. The project supports **SDG 1 – No Poverty, SDG 5 – Gender Equality, SDG 8 – Decent Work, and SDG 10 – Reduced Inequalities**.

CHAPTER 1

INTRODUCTION

Informal work plays an important role in the Indian economy. Many work as domestic helpers, beauticians, tailors, cooks, vendors, care workers, and other independent service providers. Although these workers gain valuable skills and experience, their work is often not maintained in a formal or portable digital record.

The absence of a documented work history creates several problems. A woman may have years of experience but may not have sufficient proof when applying for a better job. Similarly, irregular income that is not digitally documented may make it difficult to demonstrate financial stability when approaching formal financial institutions. Informal workers may also find it difficult to discover government welfare schemes for which they may potentially be eligible. In addition, when an employer fails to pay wages or suddenly terminates the work arrangement, the worker may have very limited documented evidence of the previous work relationship.

The proposed **Digital Work Passport** addresses this problem by creating a digital identity of work experience. The system allows workers to maintain their work history through a simple website

The application is designed with **voice interaction and regional-language support** so that users do not have to depend entirely on complicated text-based forms.

After completing a job, the employer or client can confirm the work and provide a rating. These confirmations gradually build a verified work history.

The system uses the accumulated information to create a **Trust Score** and **verified Income Record**. These records can then be used to provide additional services such as job matching, government-scheme information, group insurance, and partner financialservice opportunities.

OBJECTIVES

- To create a digital and portable record of informal work experience.
- To allow employers or clients to verify completed work.
- To maintain verified work history and income information.
- To generate a Trust Score based on verified work activity.
- To provide better job opportunities through skill-based matching.
- To provide information about potentially relevant government schemes.
- To support financial inclusion through alternative work-related information.
- To provide group insurance opportunities through suitable partners.
- To make the system accessible using voice and regional languages.

SDG ALIGNMENT

SDG 1 – No Poverty

Supports improved access to income opportunities and financial inclusion.

SDG 5 – Gender Equality

Improves the visibility and economic participation of workers.

SDG 8 – Decent Work

Supports better job discovery, work documentation, and productive employment.

SDG 10 – Reduced Inequalities

Reduces information and access barriers faced by informal workers.

CHAPTER 2
LITERATURE REVIEW

Digital platforms for informal and unorganized workers have increasingly focused on worker registration, employment support, social-security access, and financial inclusion. Existing worker-registration approaches demonstrate the importance of maintaining structured information about informal workers. However, registration alone does not necessarily provide a detailed record of the individual jobs completed by a worker.

The proposed Digital Work Passport focuses on maintaining a **verified history of actual work activities**. Instead of depending only on self-reported information, the system allows employers or clients to confirm completed jobs and provide ratings. Digital financial services have also explored alternative information for users who may not have traditional financial histories. For informal workers, verified work activity and income records can provide additional information to partner financial institutions.

Job recommendation systems use information such as skills, experience, location, and preferences to identify suitable opportunities. A similar approach can be applied to informal workers by matching their verified work profile with available jobs.

Voice-based and regional-language systems can further improve accessibility. Workers who may find conventional digital forms difficult can interact with the system through speech-assisted workflows.

CHAPTER 3

PROPOSED SYSTEM

The proposed system is a **Digital Work Passport platform** designed for informal workers.

The system starts with worker registration through a website. The application provides a simple interface with voice and regional-language support.

The worker creates a profile containing information such as occupation, skills, services offered, preferred location, and work preferences.

When the worker completes a job, the employer or client receives a request to confirm the work. The employer can verify the job and provide a rating.

The confirmed work is added to the worker's digital work history. As more jobs are completed and verified, the worker develops a stronger digital record.

The system then processes the verified information to generate a **Trust Score**. Factors such as completed verified jobs, ratings, work continuity, recent activity, and recorded income can contribute to the score.

The system also maintains an **Income Record**, where available, providing a structured view of the worker's earning history.

RESEARCH GAP

- Informal workers often lack verifiable proof of work experience.
- Traditional employment records may not represent informal jobs.
- Informal income may not be maintained in a structured digital format.
- Job opportunities and worker records are often separated.
- Government-scheme information may not be easily accessible to every worker.
- A unified platform connecting **work history, verification, trust, income, and opportunities** can address these gaps.

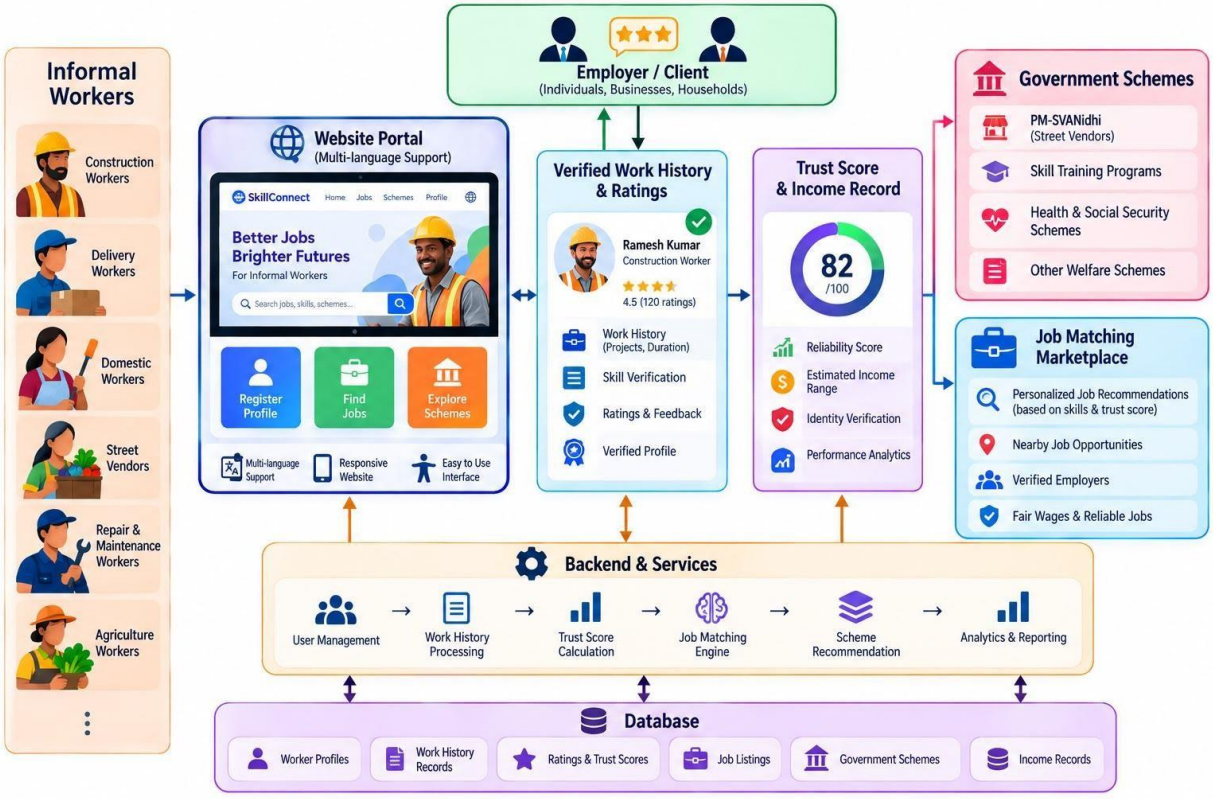
The verified profile can then be used by different services:

- Job Matching
- Government Scheme Information
- Micro-loan Partner Services
- Group Insurance
- Worker Profile Sharing

The system does not automatically approve loans, insurance, or government benefits. Instead, it provides verified information and suitable pathways, while the respective partner or authority makes the final decision.

3.1 ARCHITECTURAL DIAGRAM

The architecture of the proposed system can be represented as follows:



The application is designed with **voice-enabled and regional-language support** to make it easier for workers with limited digital literacy to use the platform. After completing a job, the employer or client can confirm the work and provide a rating. These confirmations are stored in the worker's digital profile and gradually build a verified work history.

CHAPTER 4 MODULES DESCRIPTION

MODULES DESCRIPTION

The system is divided into four major modules. Each module performs a specific function and together they form the complete Digital Work Passport system.

Module	Description
Module 1 – Worker Registration & Voice Interface	Creates the worker profile and collects skills and preferences.
Module 2 – Work Verification & Ratings	Records completed work and obtains employer/client confirmation.
Module 3 – Trust Score & Income Record	Processes verified work information and generates a trust profile.
Module 4 – Access & Recommendation Services	Provides job matching, scheme information, insurance and financial-service pathways.

4.1 MODULE 1 – WORKER REGISTRATION & VOICE INTERFACE

This module creates the digital profile of the informal woman worker.

The application is designed to be simple and accessible. Voice interaction allows the worker to provide information without depending completely on typing.

Regional-language support allows the worker to interact with the system in a familiar language.

Functions

- Worker registration.
- User authentication.
- Occupation selection.
- Skill registration.
- Service selection.
- Preferred working location.
- Work preferences.
- Voice-based interaction.
- Regional-language support.
- Consent-based data collection.

4.2 MODULE 2 – WORK VERIFICATION & RATINGS

This module records the actual work performed by the worker.

After completing a job, the employer or client can verify that the service was provided.

The employer or client can also provide a rating. This information becomes part of the worker's verified digital work history.

Functions

Create job record.

- Record job type.
- Record date of work.
- Send verification request.
- Employer/client confirmation.

- Rating collection.
- Feedback collection.
- Store verified work history.
- Detect inconsistent records.

4.3 MODULE 3 – TRUST SCORE & INCOME RECORD

This is one of the important AI/data-processing modules of the system.

The module analyzes verified work information to create a Trust Score.

The score can consider factors such as:

- Number of verified jobs.
- Rating patterns.
- Work completion.
- Work continuity.
- Recent activity.
- Recorded income.
- Consistency of work history.

The system also maintains the worker's verified income record whenever income information is available.

Functions

- Aggregate verified work records.
- Maintain income history.
- Analyze ratings.
- Analyze work continuity.

- Calculate Trust Score.
- Provide explainable score factors.
- Detect unusual activity.

4.4 MODULE 4 – ACCESS & RECOMMENDATION SERVICES

This module uses the worker's verified profile to provide useful opportunities.

The job recommendation system can match workers with suitable jobs based on:

- Skills
- Experience
- Location
- Work preferences
- Previous verified work

The system can also provide information about potentially relevant government schemes.

Partner organizations can use the verified record as an additional input for financial or insurance services, subject to their own eligibility and verification procedures.

Functions

- Job matching.
- Skill-based recommendations.
- Location-based recommendations.
- Government-scheme information.
- Micro-loan partner pathway.
- Group insurance pathway.

CHAPTER 5

IMPLEMENTATION AND RESULTS

The proposed system can be implemented as a mobile-first application connected to a secure backend.

The frontend provides separate interfaces for workers and employers/clients. The backend manages authentication, work records, verification, Trust Score calculation, recommendations, and partner-service requests.

5.1 IMPLEMENTATION SETUP **Frontend**

- Mobile/Web application
- Simple user interface
- Voice interaction
- Regional-language support
- Worker dashboard
- Employer/client verification interface **Backend**
- User authentication
- Worker profile management
- Job management
- Verification service
- Rating service
- Trust-score calculation
- Recommendation service

Database

Stores:

- Worker profiles
- Skills
- Job records
- Verification records
- Ratings
- Income records
- Trust-score information
- Consent information

AI Layer

The AI layer can be used for:

- Trust-score calculation
- Job recommendation
- Anomaly detection
- Voice processing
- Natural-language processing

Security Layer

- Authentication
- Role-based access
- Data encryption
- Consent management
- Secure information sharing
- Audit logs

PROPOSED AI TECHNIQUES

1. Trust Score Calculation

A weighted scoring approach can combine verified work activity, ratings, work continuity, and other permitted signals.

2. Job Recommendation

A recommendation algorithm can match worker skills and preferences with suitable job opportunities.

3. Anomaly Detection

AI can identify unusual patterns such as repeated suspicious ratings or inconsistent work records for further review.

4. Voice Processing

Speech-to-text and text-to-speech technologies can support voice interaction in regional languages.

5. Eligibility Filtering

A rule-based system can compare available worker information with published scheme criteria and display potentially relevant schemes.

5.2 EXPECTED RESULTS

Since this is a proposed system, these are **expected results**, not measured experimental results.

The system is expected to:

- Create a portable digital work history.
- Provide verified proof of completed work.
- Maintain a structured income record.

- Generate a Trust Score.
- Improve access to suitable job opportunities.
- Help workers discover relevant government schemes.
- Provide additional information for partner financial services.
- Support group insurance opportunities.
- Reduce dependence on informal references.
- Improve accessibility through voice and regional-language support.

EXPECTED SYSTEM OUTPUT

Worker → WEBSITE→ Job → Verification → Trust Score → Opportunities

The final system should provide the worker with a single digital profile that represents her **skills + verified experience + ratings + income history**.

CHAPTER 6
CONCLUSION AND FUTURE WORK

The Digital Work Passport for Informal Workers provides a proposed digital solution for documenting informal employment experience.

The system converts individual work activities into a structured digital work history through employer/client verification and ratings.

The accumulated information can be used to generate a Trust Score and maintain an income record. These records can then support job recommendations and provide pathways to financial, insurance, and government-scheme services.

The system combines mobile technology, artificial intelligence, voice interaction, regional-language support, verification, and recommendation techniques.

The main objective is not simply to create another worker-registration application, but to create a **portable and verifiable record of actual work performed**.

The proposed system can contribute to the broader objectives of **No Poverty, Gender Equality, Decent Work, and Reduced Inequalities**.

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